Paleomagnetism of Archean granites and Matachewan dikes in the Wawa Subprovince, Ontario: reevaluation of the Archean apparent polar wander path[1]

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Paleomagnetic measurements have been completed on 400 specimens from dated Archean granites and Matachewan dikes in the Michipicoten and Gamitagama greenstone belts in the western Wawa Subprovince of the Superior Province, Ontario. Detailed alternating-field and thermal step demagnetization analyses were used to isolate stable remanence directions. A single-component remanence was isolated within three adjacent dated granitic plutons on the eastern margin of the Michipicoten belt, including the Hawk Lake trondhjemite, the Southern external granite, and the Eastern external granite (HSE). The maximum possible age for this remanence is constrained by the intrusion of the last pluton at 2694 Ma. The corresponding HSE paleopole is located at 10°W, 41°S (dp = 8°, dm = 13°). A second paleopole, NB, is derived from the Northern external granite and the Baldhead River quartz monzonite, which give U–Pb zircon ages of 2662 and 2668 Ma, respectively. Their single-component remanence defines a paleopole on the Archean apparent polar wander path (APWP) at 15°E, 27°S (dp = 8°, dm = 13°), with a maximum possible age of 2.66 Ga. A third paleopole, GD, is derived from the north-northwest-trending Gamitagama diabase dikes and yields a position of 57°E, 41°N (dp = 7°, dm = 14°), which agrees with poles determined by other workers from the 2454 Ma Matachewan dike swarm. The GD pole, along with previously determined Matachewan dike poles, demonstrates that a tectonically stable craton has existed since intrusion of this extensive dike swarm, and it improves the precision of the 2454 Ma Matachewan pole on the APWP. These poles, when compared with coeval poles from the eastern side of the Kapuskasing Structural Zone in the Superior Province, imply no tectonic rotation or translation between the Wawa and Abitibi subprovinces along this Early Proterozoic structure.

Les mesures paléomagnétiques ont été affectuées sur 400 échantillons de granites archéens et de dykes de Matachewan datés dans les ceintures de roches vertes de Michipicoten et de Gamitagama, de la sous-province de Wawa accidentale dans la province du Supérieur, en Ontario. Une désaimantation en échelon détaillée par champ alternatif et par chauffage a permis d'isoler les directions de rémanence stable. Une rémanence à une composante a été isolée dans trois plutons granitiques adjacents d'âge connu, localisés sur la bordure orientale de la ceinture de Michipicoten et incluant la trondhjémite d'Hawk Lake, le granite externe Sud et le granite externe Est (HSE). Le dernier pluton limite l'âge maximum possible et cette rémanence à 2694 Ma. Le paléopôle HSE correspondant est localisé à 10°W, 41°S (dp = 8°, dm = 13°). Un second paléopôle, NB, est dérivé du granite externe Nord et de la monzonite quartzique de Baldhead River qui ont fourni des âges U–Pb sur zircon de 2662 et 2668 Ma, respectivement. Leur rémanence à une composante définit un paléopôle sur la courbe de dérive apparente des pôles de l'Archéen à 15°E, 27°S (dp = 8°, dm = 13°), avec un âge maximum possible de 2,66 Ga. Un troisième paléopôle, GD, est dérivé des dykes de diabase de Gamitagama, de direction nord-nord-ouest, et fournit une position de 57°E, 41°N (dp = 7°, dm = 14°), en accord avec les pôles déterminés par d'autres travaux sur l'essaim de dykes de Matachewan âgé de 2454 Ma. Le pôle GD, couplé à d'autres pôles de dykes et Matachewan déterminés antérieurement, démontrent qu'un craton tectoniquement stable existait depuis l'intrusion de ce vaste essaim de dykes, et il apporte des précisions quant à la position du pôle de Matachewan de 2454 Ma sur la courbe de la dérive apparente des pôles. Ces pôles, comparés avec d'autres pôles du même âge de la bordure orientale de la Zone structurale de Kapuskasing dans la province du Supérieur, impliquent l'absence de rotation tectonique ou de translation entre les sous-provinces de Wawa et d'Abitibi le long de cette structure d'âge du Protérozoïque précoce.
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Introduction

To date there are relatively few paleopoles determined for Archean rock units in the North American Shield, and most of these are poorly dated. As a result, the acquisition times for their remanent magnetizations are even more poorly known. Thus, the Archean apparent polar wander paths (APWP) that have been constructed for North America (e.g., Irving and Naldrett 1977; Irving 1979; Seguin *et al.* 1983; Dunlop 1984*a*) are for the most part based on paleopoles with large error limits on their ages. This age problem was demonstrated most recently when a more precise U–Pb zircon age for the Matachewan–Hearst dikes of 2454 ± 2 Ma was determined (Heaman 1988), replacing the previous Rb–Sr isochron age of 2633 ± 91 Ma (Gates and Hurley 1973; Easton 1986).

In this paper we describe new paleomagnetic results from U–Pb-dated granites and dikes in the Wawa Subprovince of the Superior Province. Our purpose was to define paleopoles from the Wawa Subprovince for the APWP and thereby see if there has been significant post-Archean translation and (or) rotation on the Kapuskasing Structural Zone (KSZ), as has been postulated (e.g., Bennett *et al.* 1967; Watson 1980; Ernst and Halls 1984).

Geology

The Michipicoten and Gamitagama greenstone belts are located in the western part of the volcanic-rich Wawa Subprovince of the Archean Superior Province. They have experienced greenschist-grade metamorphism and are separated by an intervening granitic terrane. Both are typical Archean greenstone belts, being comprised mostly of mafic to felsic metavolcanic flows and pyroclastics with interbedded metasediments (Ayres 1969; Sage 1979; Attoh 1980). These strata have been intruded by a number of granitic and gabbroic dikes, sills, stocks, and plutons that are likely subjacent intrusive equivalents of the extrusive rocks (Ayres 1969). Surrounding the greenstone belts are regional masses of granite and granitic gneiss that make up the external granitic terrane. From U–Pb

[1]Lithoprobe Publication 162 from the Kapuskasing Structural Zone transect.

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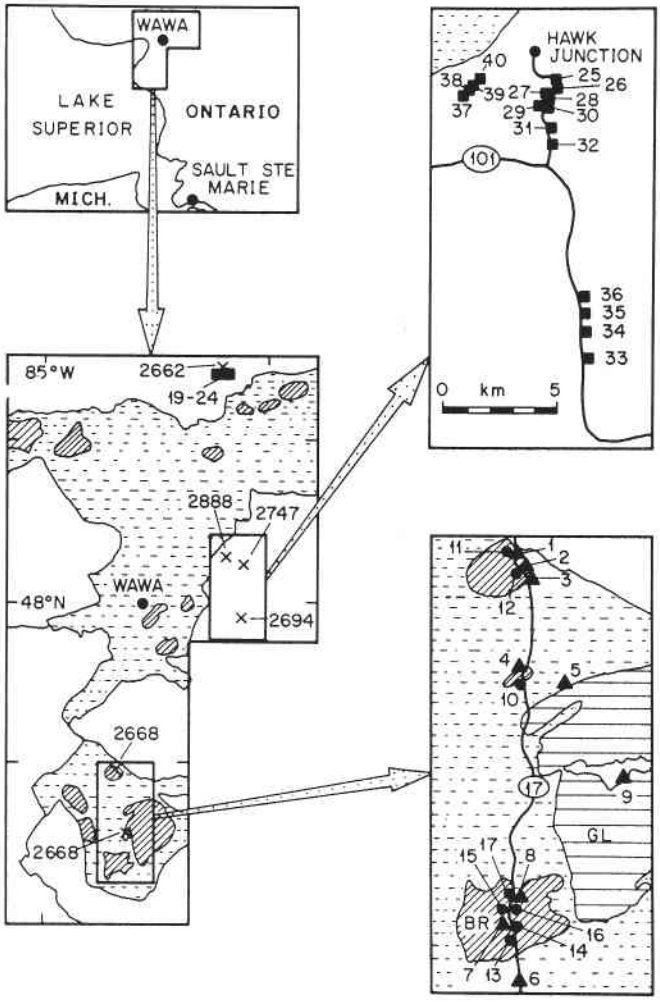


FIG. 1. Michipicoten and Gamitagama greenstone belts with their age date localities (after Turek *et al.* 1984) and the paleomagnetic sites. The Northern external granite sites (19–24) are shown by the closed rectangle and the Baldhead River quartz monzonite sites by the closed circles (NB); the combined Hawk Lake trondhjemite, Southern external granite, and Eastern external granite sites (HSE), by the closed squares; and the Gamitagama dike sites (GD), by the closed triangles. BR, Baldhead River quartz monzonite; GL, Gamitagama Lake complex. Both BR and GL are enclosed by greenstones.

zircon geochronology Turek *et al.* (1984) determined that the volcanism in these belts extended from at least 2749 to 2696 Ma and that the plutonism extended from at least 2888 to 2615 Ma. Transecting all older rock units are the Matachewan and the Abitibi dike swarms, which trend north-northwest and northeast, respectively.

Methods

The external granitic terrane was sampled in three abutting plutons on the east side of the Michipicoten Belt, namely, the Hawk Lake trondhjemite (2747 ± 7 Ma), the Southern external granite (2694 ± 3 Ma), and the Eastern external granite (2888 ± 2 Ma) (Turek *et al.* 1982, 1984) (HSE sites 25–40; Fig. 1). Also sampled were the Northern external granite (2662 ± 5 Ma) and the Baldhead River quartz monzonite (2668 ± 2 Ma) (NB sites 10–24) and the north-northwest-trending Gamitagama diabase dikes, which are part of the Matachewan dike swarm (2454 ± 2 Ma) (Krogh and Turek 1982; Turek *et al.* 1984; Heaman 1988) (GD sites 1–9; Fig. 1).

At least five core samples were drilled and oriented *in situ* by magnetic and solar compass at each of the 40 sites. Two specimens were cut from each core. The specimens were analyzed inside a magnetically shielded room using a Schonstedt SSM-1A spinner magnetometer, CTF cryogenic magnetometer, Schonstedt GSD-1 AF demagnetizer, and Schonstedt TSD-1 thermal demagnetizer. After measuring the natural remanent magnetization (NRM), two independently oriented specimens per site were demagnetized in 11 or 12 steps (selected from steps 5, 10, 15, 20, 21, 27, 30, 35, 40, 45, 50, 55, 60, 70, 85, and 99; units in mT) by alternating fields (AF's) to 99 mT. A further two independently oriented specimens per site were thermally demagnetized in 11 or 12 steps (selected from steps 22, 100, 200, 300, 400, 450, 500, 525, 545, 575, 600, 625, 640, and 650; units in °C) to 650°C. Orthogonal vector decay plots indicated that single stable components were isolated after demagnetization removed the "soft" remanence residing in low-coercivity, low-laboratory-unblocking-temperature domains. Directions were obtained by a joint analysis of stereographic plots, orthogonal demagnetization diagrams, and principal-component analysis (Kirschvink 1980). Directions of remanence were accepted with a mean angular deviation (MAD) $<10°$. The remaining specimens from each site were demagnetized using a combination of four AF steps and one thermal step, which varied for each site depending on the detailed step demagnetization behavior (selected from AF steps 10, 15, 20, 21, 27, 30, 35, 55, 70, 85, and 99 (units in mT) and thermal step 500°C). Site mean remanence directions were calculated using conventional tiered statistics (Fisher 1953; Irving 1964).

Ten specimens, representing each of the major lithologies sampled, were subjected to saturation isothermal remanent magnetization (SIRM) analysis to characterize their magnetic remanence carriers.

Results

All accepted specimen directions displayed single-component demagnetization vectors after the initial removal of soft viscous remanence components by about 15 mT or 300°C. Orthogonal vector decay plots are shown in Fig. 2 for each of the rock units in which its characteristic remanence direction is isolated. For both the HSE and NB group sites, antiparallel polarity directions were isolated (Figs. 2*b*, 2*d*–2*f*). Alternating-field demagnetization produced better results for all of the analyzed rock units and isolated their characteristic remanences in domains with coercivities up to 99 mT (Figs. 2*b*–2*d*, 2*f*). Examples of stable thermal step-demagnetized specimen directions from units NB and HSE are shown in Figs. 2*a* and 2*e*, respectively. However, the external granites in general and the diabase dikes in particular did not respond well to thermal demagnetization, which isolated the same components but in fewer specimens and with poor precision. As is often found in Archean rocks, especially granites, many of the specimens displayed erratic unstable remanence directions on demagnetization. The rejection of these data from these specimens led to low N values for several sites in Table 1. Site mean directions with α_{95}'s $> 35°$ were considered unstable and, therefore, were rejected from unit mean calculations. One could argue for a more conservative site mean acceptance criteria of α_{95}'s $< 20°$. However, the unit mean directions would change by negligeable amounts except for a small decrease in precision because of the smaller numbers of sites. In addition, all accepted site mean directions were tested for randomness by comparing the calculated R value with the tabulated significance point R_o for different values of N (Irving 1964). This test demonstrated that our accepted site

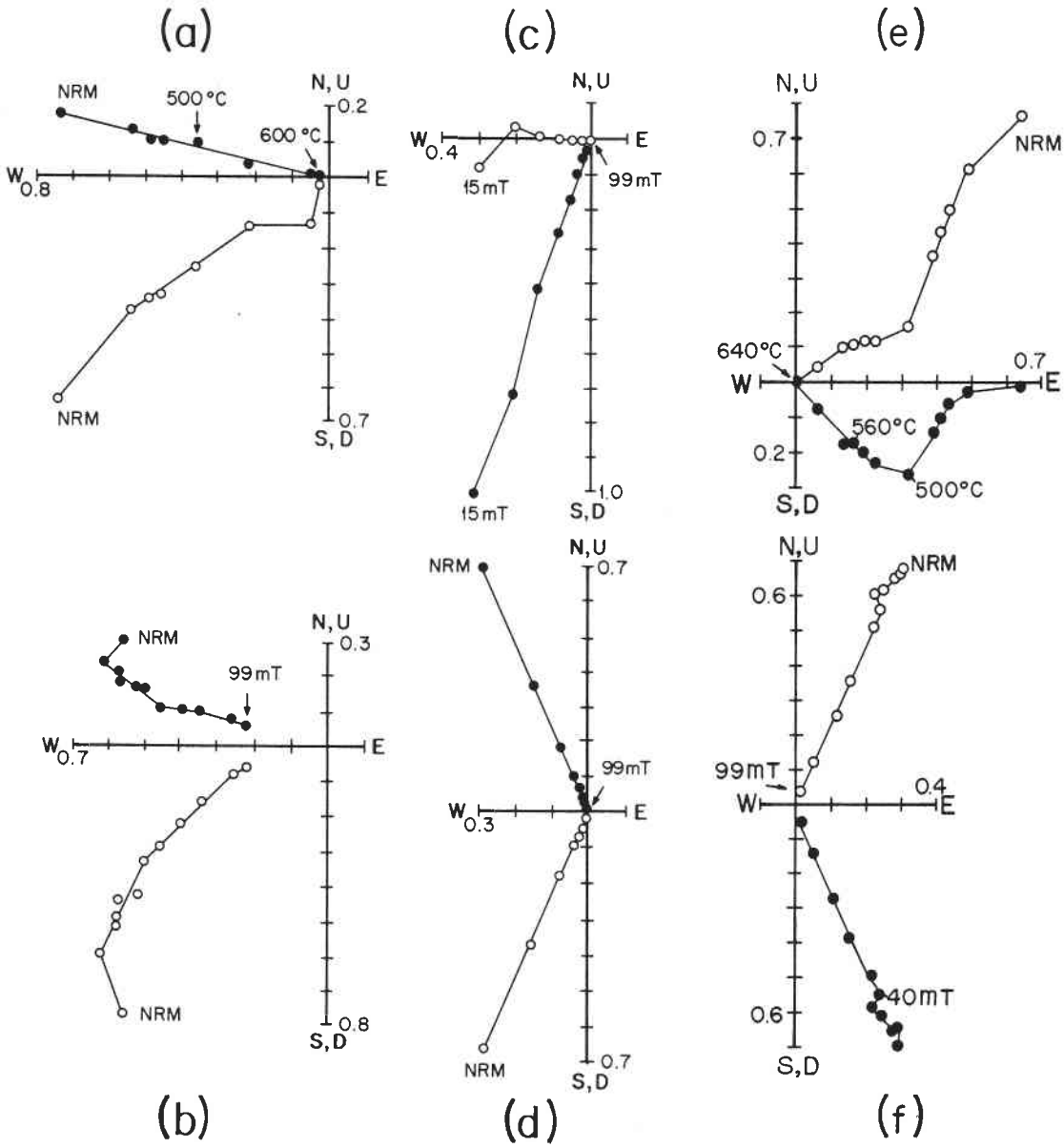


FIG. 2. Example orthogonal vector decay plots on stepwise demagnetization. Closed symbols are projections on the horizontal plane, open symbols projections on the vertical plane. The axial values are given as a ratio of the NRM intensity: (*a*) thermal pilot from site 11 (NB); (*b*) AF pilot from site 11 (NB); (*c*) AF pilot from diabase dike site 4 (GD); (*d*) AF pilot from site 36 in the Southern external granite (HSE); (*e*) negative polarity thermal pilot from site 20 (NB); and (*f*) negative polarity AF pilot from site 36 (HSE).

means have a greater than 95% probability that their directions have not arisen by chance and are therefore significant for directional studies of the paleomagnetic field. Although the data from site 5 pass this test, they were rejected because their remanence directions were best grouped at only 10 mT and thus were likely to have retained undemagnetized viscous components. Thus, only those sites with α_{95}'s $< 35°$, excluding site 5, were used to calculate the three unit means (Table 2). Although some of the site mean dispersions are relatively high in the granite rocks, this is expected because these are relatively coarse-grained rocks that contain a smaller percentage of ferromagnetic minerals.

Both magnetic polarities were observed in all of the sampled rock units except for the diabase dikes, which retained only one polarity. By convention, all specimen directions with negative inclinations were converted to their antipodal positive directions to calculate the unit means except for the diabase dikes (Fig. 3). Each unit shows a good cluster of site means, with a degree of directional dispersion that can be reasonably attributed to secular variation.

It is important to observe that the site mean directions of the granitic rocks—in particular the NB group, which includes data from the Baldhead River quartz monzonite (BRQM)—are very different from the diabase dike site means (Fig. 3). Seven of the nine sampled diabase dikes physically intrude the BRQM, which yields a markedly different remanence direction, providing a positive contact test (Fig. 1). Therefore, it is unlikely that a post-Matachewan remagnetization event could have reset the magnetization of these granitic rocks without also having reset the Matachewan dike remanence. Thus, in all probability the magnetizations from the granitic rocks predate the Matachewan dike remanence.

The SIRM acquisition and demagnetization curves for the six HSE specimens show a variety of behaviors (Fig. 4*a*). Speci-

TABLE 1. Site mean remanence data

	Unit	N	n	AF (mT)	Polarity[a]	D (°)	I (°)	R	k	α_{95} (°)
1	GD	5	0	35	n	221.6	46.5	1.817	1	61.4*
2	GD	5	0	55	r	202.9	−16.1	4.800	20	14.0
3	GD	5	0	99	n	150.7	12.7	2.229	1	67.2*
4	GD	4	1	99	2/2	192.2	00.4	3.393	5	31.3
5	GD	5	0	10	3/2	243.9	09.0	4.799	20	17.6*
6	GD	5	0	27	r	219.0	−09.7	4.993	565	3.2
7	GD	5	0	70	r	218.3	−16.7	4.860	29	11.6
8	GD	5	0	35	r	221.2	−20.1	4.982	222	5.1
9	GD	5	0	85	2/3	193.6	03.7	4.542	9	20.9
10	NB	5	0	15	n	295.0	63.2	4.957	93	7.9
11	NB	4	0	20	n	285.9	30.2	3.890	27	13.4
12	NB	4	1	20	n	281.7	52.8	3.770	13	19.4
13	NB	4	0	55	2/2	275.4	37.1	3.655	9	23.3
14	NB	4	1	55	n	068.7	68.1	2.242	2	49.5*
15	NB	4	0	27	3/1	277.7	68.1	3.952	63	8.8
16	NB	3	0	99	n	7.1	32.8	2.233	3	78.9*
17	NB	5	0	10	n	124.0	49.0	4.000	4	41.5*
18	NB	4	1	55	n	298.3	40.1	3.810	16	17.5
19	NB	5	0	30	r	082.8	−35.1	4.784	19	14.5
20	NB	6	1	21	r	102.9	−32.8	5.440	9	19.1
21	NB	5	0	21	3/2	65.8	15.9	4.010	4	41.5*
22	NB	5	0	21	2/3	290.0	35.9	4.364	6	25.1
23	NB	3	2	21	n	106.5	13.6	0.606	1	80.8*
24	NB	6	1	21	n	20.1	53.9	4.644	4	40.5*
25	HSE	4	1	30	n	307.5	41.1	3.752	12	20.1
26	HSE	5	0	30	n	324.0	55.1	4.656	12	18.4
27	HSE	4	1	30	n	289.8	54.2	3.830	18	16.7
28	HSE	3	2	21	2/1	277.3	36.4	2.852	14	22.0
29	HSE	5	0	30	n	86.5	74.4	2.159	1	57.8*
30	HSE	5	0	21	2/3	302.2	52.4	4.550	9	21.0
31	HSE	2	3	30	n	58.8	16.8	1.819	5	—*
32	HSE	4	1	21	n	239.2	28.5	3.334	4	35.0*
33	HSE	4	1	21	n	22.2	33.2	2.835	3	62.8*
34	HSE	4	1	30	n	134.8	6.1	3.278	4	36.0*
35	HSE	3	2	30	n	304.6	26.8	2.889	18	19.1
36	HSE	3	2	21	2/1	312.0	17.8	2.803	10	25.5
37	HSE	4	1	21	n	321.0	40.4	3.671	9	23.2
38	HSE	2	3	21	n	315.1	27.4	1.993	135	8.5
39	HSE	4	1	30	3/1	328.1	20.3	3.586	7	28.4
40	HSE	1	4	21	n	305.4	29.0	—	—	—*

NOTES: Unit abbreviations are as follows: GD, Gamitagama diabase dikes; NB, Northern external granite and Baldhead River quartz monzonite; HSE, Hawk Lake trondhjemite, Southern external granite, and the Eastern external granite. *N*, number of core means averaged; *n*, number of cores rejected; AF, alternating-field demagnetization step; *D* and *I*, declination and inclination; *R*, resultant vector; *k*, precision parameter; α_{95}, radius of 95% confidence (Fisher 1953). Asterisks denote rejected sites.

[a]Polarity of the site is given as normal (n), reverse (r), or mixed, in which case the ratio of normal to reverse polarity cores is indicated.

men *a* shows only multidomain (MD) magnetite behavior. Specimens *b–d* show the rapid initial remanence acquisition characteristic of some MD magnetite content, with a subsequent slower rate of acquisition that is indicative of predominantly pseudosingle-domain (PSD) to single-domain (SD) magnetite. In addition, specimen *d* appears to have a significant hematite content, denoted by the slow continued remanence acquisition above 0.3 T. Specimens *e* and *f* have PSD to SD magnetite behavior and also appear to have some hematite content. Both specimens *g* and *h* from the NB granite group have curves that indicate the dominance of PSD to SD magnetite, with specimen *g* having a little MD magnetite also (Fig. 4*b*). Finally, specimens *i* and *j* from the GD diabase dike group both show typical MD magnetite behavior only. As expected from samples of different lithologies, there are a variety of magnetic carriers within the sampled units. In general, the sites with the greatest dispersion in remanence direction, as measured by larger α_{95}'s (Table 1), are also those with the most variable combination of carriers.

Discussion

The Archean APWP for North America, as pointed out by Roy (1983), is based on a nonsystematic sampling of a noncontinuous paleomagnetic record. This is a fundamental problem

TABLE 2. Unit means and their paleopoles

Unit	N	D (°)	I (°)	R	k	α_{95} (°)	ϕ (°)	λ (°)	dp (°)	dm (°)
GD	6	207.7	−10.0	5.798	25	13.7	41N	57E	7	14
NM	9	283.0	44.3	8.692	26	10.3	27S	15E	8	13
HSE	10	308.9	38.1	9.537	20	11.2	41S	10W	8	13

NOTES: N, number of site means averaged; dp, radius of the ellipse of 95% confidence at the pole along the site-pole great circle; dm, radius of this ellipse perpendicular to the site-pole great circle; λ, longitude; ϕ, latitude; D, I, R, k, and α_{95} as in Table 1.

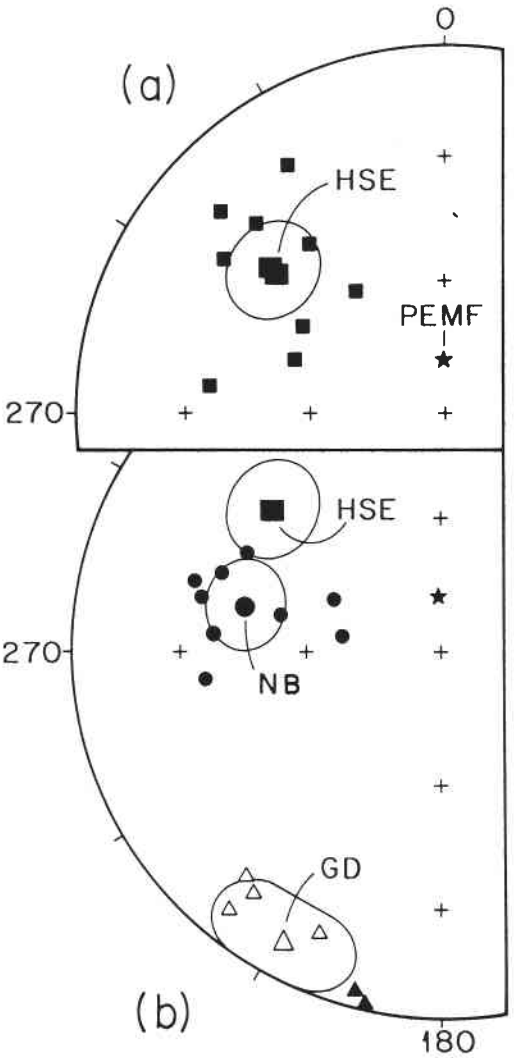


FIG. 3. Lower hemisphere stereographic projections of site mean remanence directions. Closed symbols are of positive (down) polarity, open symbols of negative (up) polarity. (*a*) HSE site means and unit mean, with the present Earth's magnetic field (PEMF). (*b*) NB and GD site means and their unit means, including the HSE unit mean. Note that unit means are illustrated as larger symbols with their corresponding α_{95}'s, and HSE and NB magnetizations are characterized by mixed polarities (Table 1); thus their negative polarities have been converted to their antipodal positive direction.

that paleomagnetists can do little about, especially in the Precambrian. Roy (1983) also noted that the magnetization age of only a few poles was known to within ±50 Ma and that for many the error was more than ±100 Ma. Thus the Archean APWP's that have been constructed for North America by Irving and Naldrett (1977), Irving (1979), Seguin *et al.* (1983), and Dunlop (1984*a*) are based for the most part on paleopoles with large error limits on their ages. Recently, the U–Pb zircon dating method has contributed much more precise age dates for many rock units in the Superior Province, in turn establishing maximum ages for the magnetizations of many Archean poles, including those determined in this study.

Also, there are tectonic problems. Hoffman (1988) has provided a model for the assembly of the Archean cratons that form the Precambrian Shield of North America. In his model the Archean Slave and Superior cratons are united into one plate during the Proterozoic by collision to form the intervening trans-Hudson orogen. Dunsmore and Symons (1990) argued from paleomagnetic data that the intervening ocean was about 4000 km wide and it closed in the formation of the orogen, and thus they have confirmed earlier suggestions that the Slave and Superior provinces require separate APWP's before 1750 Ma (e.g., Cavenaugh and Seyfert 1977; Irving *et al.* 1984). For this reason, poles such as the Easter Island and Dogrib dikes poles from the Slave Province (Irving *et al.* 1984) and the Stillwater Complex pole from the Wyoming Province (Geissman and Harlan 1988) are omitted from our evaluation, and only poles from the Superior Province itself are considered.

The Matachewan dikes, dated by Heaman (1988) at 2454 ± 2 Ma, yield the most extensively studied and well-defined dual-polarity Archean remanence in the Superior Province. Poles determined for the dikes by Irving and Naldrett (1977), Shutts and Dunlop (1981), Geissman *et al.* (1983), Ernst and Halls (1984), and Halls and Shaw (1988) are all in close agreement with the GD pole of this study. This agreement indicates that a tectonically stable craton has existed since emplacement of this extensive dike swarm and that the postintrusion rotation of about 15° between the subterranes across the KSZ, as postulated by Bennett *et al.* (1967), is unlikely. Further, the Matachewan poles provided the 2.63 Ga position on the previously published APWP's, and it was considered the best-defined Archean result. Revision of this position to an age of no more than about 2.45 Ga leads to a major revision of the APWP (Fig. 5; Table 3).

The Matachewan dikes in the Gamitagama greenstone belt that were sampled in this study cut the BRQM and have the typical north-northwest Matachewan trend and paleomagnetic signature that provides a positive contact test. Further, Matachewan dikes are known to cut all of the other rock units sampled in this study (Ayres 1969; Ernst and Halls 1984), and therefore the HSE and NB magnetizations in all probability are older than 2.45 Ga. Given that the BRQM and the Northern external granite have essentially the same remanence direction and U–Pb zircon age, and that this direction fits concordantly on the Archean APWP with poles determined from rocks of similar age for the Superior Province, it follows that the age of this magnetization is >2.45 Ga, the time of Matachewan dike intrusion, but <2.66 Ga, which is the U–Pb age of the NB group granites. As there is no evidence of a later postintrusive

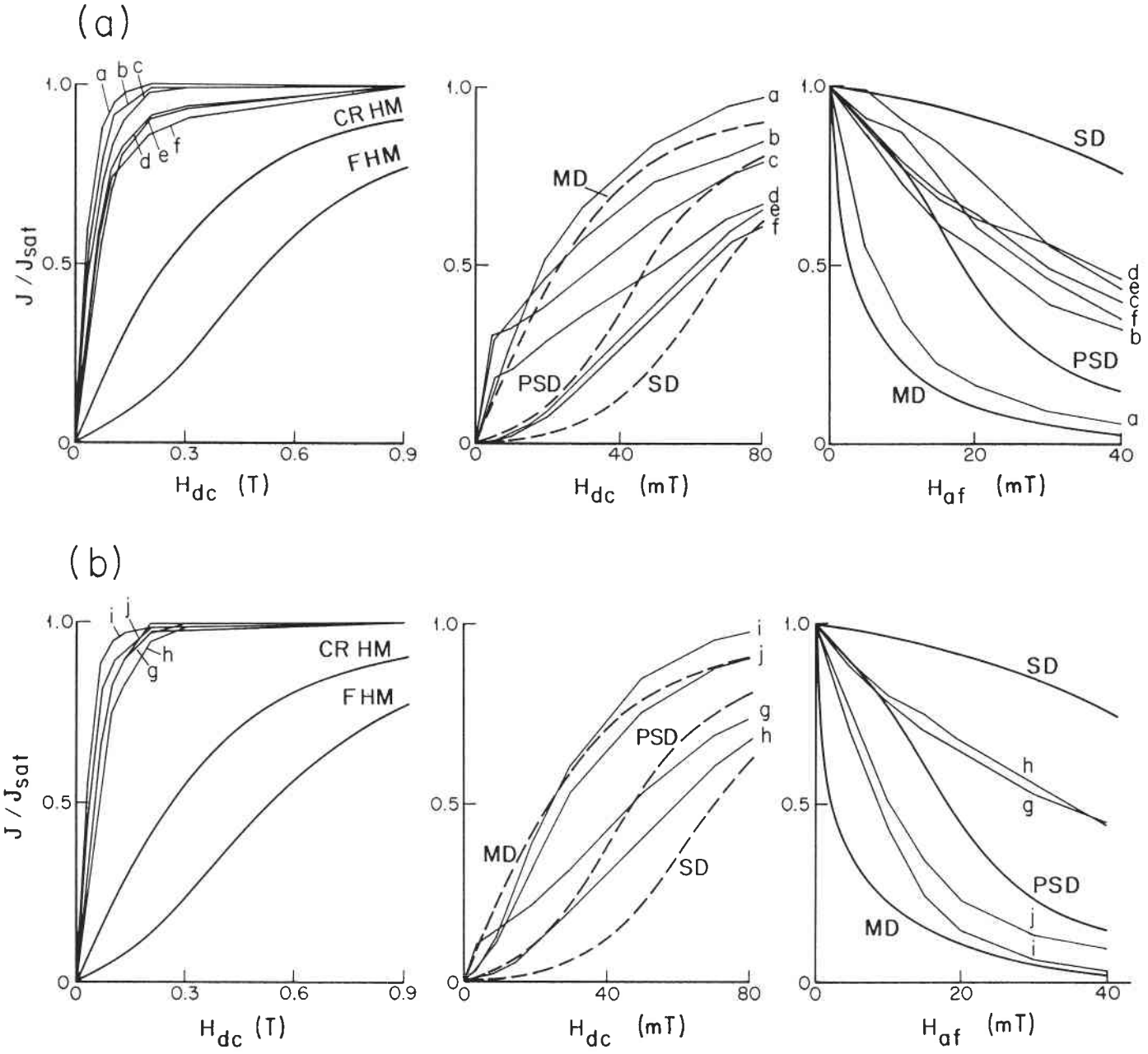


FIG. 4. SIRM acquisition and demagnetization curves for example specimens from the (*a*) HSE group of sites and (*b*) NB (specimens *g* and *h*) and GD (specimens *i* and *j*) groups of sites. Centre row figures are exploded views of SIRM acquisition curves near the origin. The type curves for single, pseudosingle, and multidomain magnetite (SD, PSD, MD) and for fine and coarse hematite (FHM, CRHM) are from Dunlop (1971, 1972, 1973, 1981).

pre-Matachewan thermal remagnetization event, we interpret the age of this magnetization as being probably close to the maximum U–Pb age of 2.66 Ga (NB; Fig. 5). A similar argument holds for the HSE remanence. The three granitic bodies that contribute to the HSE pole are physically adjacent and have the same magnetization direction, and their pole also fits concordantly on the APWP. Thus, we interpret the age of this remanence as being probably close to that of the latest intrusive phase, which is the Southern external granite dated at 2694 Ma (Turek *et al.* 1984) (Fig. 5). This argument for the age of magnetization requires that there was no postemplacement tilt in the area, which fits with the evidence. Percival and Card (1983) argued from regional metamorphic grades that the Wawa area could not have been tilted by more than a very few degrees. Also, the Matachewan dikes are near vertical in the study area, which suggests no postemplacement tilt.

Across the KSZ to the east in Abitibi Subprovince, Jensen and Langford (1983) defined four volcanic cycles that the U–Pb zircon dates by Nunes and Jensen (1980), Nunes and Pyke (1980), and Corfu *et al.* (1989) place between 2698 and 2747 Ma. Thus the maximum age for (*i*) the Skead volcanics pole (SK) is 2701 ± 3 Ma; (*ii*) the Archean basalts and gabbros (BG), Kinojevis volcanics (KJ), Ghost Range Complex (GR), Adams Mine volcanics and iron formation (AMA), and Dundonald sill (DS) poles is 2706 ± 6 Ma; and (*iii*) the Sherman Mines (SMA) and Moose Mountain Mine (MMA) volcanics and iron formation poles is 2722 ± 20 Ma. Given these maximum ages, it is clear that the 2700 Ma positions shown on

TABLE 3. List of accepted Archean Superior Province paleopoles

Pole	Unit means D (°)	I (°)	k	α_{95} (°)	N	ϕ (°)	λ (°)	dp (°)	dm (°)	Max. age (Ma)
MD[a]	207.0	−16.0	54	7.0	10	44N	60E	5.0[d]		2454±2
MD2[a]	021.0	13.0	44	6.0	14	45N	70E	—	—	2454±2
MD3	019.4	20.2	40	6.6	12	49N	69E	3.6	6.9	2454±2
MD4	204.0	−11.6	52	5.4	17	44N	62E	2.8	5.4	2454±2
KJ[a]	194.9	−14.3	8	7.9	38[c]	47N	78E	4.1	8.1	2454±2
GD[b]	207.7	−10.0	25	13.7	6	41N	57E	14.0	7.0	2454±2
SL1	005.9	56.4	95	4.4	12	78N	66E	4.7	6.4	2580±20
PLN	359.5	55.5	28	8.3	13	78N	90E	8.5	12.0	2700±25
BL	002.3	48.9	24	10.0	10	71N	83E	10.3	14.2	
CS	006.0	38.0	70		4	61N	93E	8.0	13.0	
PLR	198.0	−22.5	62	5.3	13	51N	60E	3.0	5.5	2700±25
SK[a]	218.0	19.0	12	8.0	28	23N	59E	4.4	8.4	2701±3
GMA	211.1	32.6	20	3.2	120[c]	14N	56E	4.0	2.0	
WB	246.0	12.0	20	10.5	11	11N	20E	5.3	10.6	
MMA[a]	080.6	6.1	11	6.4	52[c]	8N	16E	8.0	4.0	2722±20
AMA[a]	256.0	7.4	22	3.3	151[c]	6N	22E	3.0	2.0	2710±2
SMA[a]	096.0	−7.0	10	7.0	70[c]	6S	8E	9.0	5.0	2722±20
GR[a]	280.0	2.0	6	11.8	33[c]	7S	13E			2706±6
DS[a]	269.0	32.0	19	18.0	5	13S	21E	19.0[d]		2706±6
KK[a]	289.0	23.0	144	5.0	7	21S	2E	5.0[d]		
NB[b]	283.0	44.3	26	10.3	9	27S	15E	13.0	8.0	2662±5
BG[a]	114.0	−32.0	15	11.0	13	29S	4E			2706±6
HSE[b]	308.9	38.1	20	11.2	10	41S	10W	13.0	8.0	2694±3

NOTES: D, I, k, α_{95}, N, ϕ, λ, dp, and dm as in Table 2. See text for a discussion of age dates. Poles are as follows: MD, Matachewan dikes; DS, Dundonald Sill; KK, Kamiskotia Complex (Irving and Naldrett 1977). MD2, Matachewan dikes; BG, basalts and gabbros (Shutts and Dunlop 1981). MD3, Matachewan–Hearst dikes (Ernst and Halls 1984). MD4, Matachewan dikes (Halls and Shaw 1988). KJ, Kinojevis metavolcanics (Geissman *et al.* 1983). GD, Gamitagama Matachewan dikes (this study). SL1, Shelley Lake granite (Dunlop 1984*a*; age from Berger and York 1979). PLN, Poohbah Lake Complex normal; PLR, Poohbah Lake Complex reverse (Dunlop 1985; age from Mitchell 1976). BL, Burchell Lake granite (Dunlop 1984*b*). CS, Chibougamau sills (Ueno and Irving 1976). SK, Skead volcanics (Ridler and Foster 1981). GMA, Griffith Mine A component (Symons *et al.* 1983). WB, Wabigoon gabbro (Dunlop 1983). MMA, Moose Mountain Mine A component (Symons *et al.* 1980). AMA, Adams Mine A component (Symons *et al.* 1981). SMA, Sherman Mine A component (Symons and Stupavsky 1983). GR, Ghost Range Complex (Geissman *et al.* 1982). NB, Northern external granite and Baldhead River quartz monzonite; HSE, Hawk Lake trondhjemite, Southern external granite, and Eastern external granite (this study).

[a]Poles from eastern side of KSZ.

[b]Poles from this study.

[c]N is number of samples.

[d]Value for A_{95}.

the earlier Archean APWP's must, in fact, be 50–100 Ma younger.

Many units providing Archean poles for the Abitibi Subprovince have been intruded by undeformed plutons such as the Otto Stock and a variety of porphyries that give U–Pb zircon ages of 2673^{+6}_{-2} to 2691 ± 3 Ma (Corfu *et al.* 1989), thereby providing one indication of their minimum age of magnetization. In addition, the age date compilation by Easton (1986) shows that many whole-rock Rb–Sr isochrons, K–Ar, mineral ages, and $^{40}Ar/^{39}Ar$ plateau ages from these units and later undeformed intrusives range from 2450 to 2650 Ma, with several ages on crosscutting intrusives of about 2550 Ma. Since these ages likely correspond to uplift, they provide a lower limit to the ages of magnetizations. All of these units are also cut by the very extensive 2454 Ma Matachewan dike swarm, and some positive contact tests indicate the host-rock remanence predates their emplacement (Irving and Naldrett 1977). Therefore, it is likely that the GR, BG, KK, DS, SMA, MMA, and AMA cluster of poles (Fig. 5; Table 3), which includes poles from prefolding remanences, is older than about 2550 Ma and less than the maximum possible age of magnetization from the high-precision U–Pb zircon age dates of about 2.7 Ga. Therefore, it is reasonable to assign a maximum age of 2.66 Ga to this portion of the APWP (Fig. 5), based on the agreement of the NB pole from this study with the cluster poles from the Abitibi Subprovince.

Finally, when the HSE, NB, and GD paleopoles determined for the Wawa Subprovince in this study are plotted with coeval paleopoles from the Superior Province on both sides of the KSZ, they appear to be concordant (Fig. 5). Thus, the paleomagnetic data to date provide no evidence to indicate any significant displacement on the KSZ during the Archean or thereafter. However, one must caution that the Archean paleomagnetic record for the Superior Province is still sketchy.

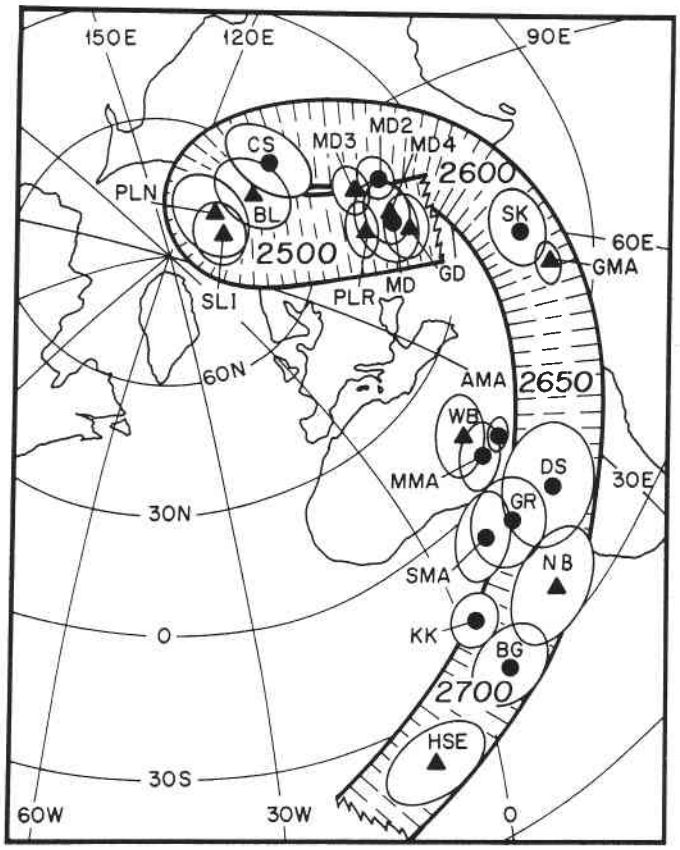


FIG. 5. Archean APWP for Superior Province of North America. New poles from this study and their corresponding maximum possible ages of magnetization: GD (2.45 Ga), NB (2.66 Ga), and HSE (2.70 Ga). Circles and triangles are poles determined from the east and west side of the Kapuskasing Structural Zone, respectively. See Table 3 for pole references.

Conclusions

Since the NB and HSE magnetizations are characterized by both normal and reverse polarity, this suggests that primary remanences have been recorded. The well-defined and characteristic Matachewan remanence found in the north-northwest-trending Gamitagama diabase dikes (GD) constrains the minimum age of the NB and HSE magnetizations at >2.45 Ga. The results from other paleomagnetic studies of the Matachewan swarm similarly constrain the minimum age for the rest of the prefolding Archean poles in the Superior Province. The presence of both polarities, a positive contact test, precise U–Pb zircon ages, and the concordancy of the NB and HSE poles with the Archean APWP derived from other rock units in other subprovinces of the Superior Province are strong arguments in favor of our interpretation of the ages of the NB and HSE remanences.

The NB, HSE, and GD paleopoles from this study, combined with the revised age for the Matachewan dikes, require revision of the APWP for the Superior Province of North America. This analysis suggests that no paleomagnetically significant tectonic translations or rotations have taken place between the Abitibi and Wawa subprovinces on the KSZ since the time of Matachewan dike intrusion and probably earlier, thus implying that a stable craton has existed since the Matachewan dike intrusion. Archean apparent polar wander for North America appears to have taken place at a very rapid rate, combined with small changes in paleolatitudes, suggesting that large-scale rotations of the Superior craton took place during the late Archean.

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